MATE70020 Problem Set 2

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Method

All calculations used `pw.x` and `bands.x` (Quantum ESPRESSO 7.6 [3]) with the supplied LDA (Perdew–Zunger) pseudopotentials `B.pz-vbc.UPF` and `N.pz-vbc.UPF`. The SCF calculation used `BN.in` unchanged: 60 Ry cutoff, $a = 2.504$ Å, $c = 20.0$ Å, $6 \times 6 \times 1$ k-grid, `nbnd = 8`. It converged in 7 iterations (estimated accuracy 3×10^{-7} Ry, below the default threshold of 10^{-6} Ry), with $E_{\mathrm{tot}} = -25.774\,88$ Ry. The band structure was then computed non-self-consistently on this density. Only two changes were made to `BN.in`: `calculation = 'bands'`, and `K_POINTS crystal_b` with the path $\Gamma\,(0,0,0)$ 60, $K\,(\frac{1}{3},\frac{1}{3},0)$ 30, $M\,(\frac{1}{2},0,0)$ 1. This gives 91 k-points with equal spacing 0.028/Å, because $|\Gamma K| = 2|KM|$. The result was post-processed with `bands.x`.[1]

(a) Why the $6 \times 6 \times 1$ grid has no k-points along z

Quantum ESPRESSO applies periodic boundary conditions in all three directions, so `BN.in` repeats the single hBN sheet along z every $c = 20.0$ Å ($\vec{a}_3 = c\,\hat{\vec{z}}$). Each sheet is separated from its artificial periodic images by about 20 Å of vacuum, like the isolated N_2 molecule in a box in Lecture 3. The Kohn–Sham orbitals of neighbouring sheets do not overlap across this vacuum, so the occupied KS energies $\varepsilon_{n\vec{k}}$ do not depend on k_z. Every k_z therefore contributes equally to the density $n(\vec{r}) = \sum_n \sum_{\vec{k}} |\psi_{n\vec{k}}(\vec{r})|^2$ (Lecture 4), and the single value $k_z = 0$ samples the zone exactly along z. The k-grid only has to be converged in the plane of the sheet, where the material is really periodic. Equivalently, the zone is very thin along z: $|\vec{b}_3| = 2\pi/c = 0.314$/Å, 9.2 times shorter than $|\vec{b}_1| = 2.897$/Å (Table 1). As a check, the occupied KS energies at $k_z = \pi/c$ (at Γ and K) differ from those at $k_z = 0$ by less than 0.01 eV, whereas band 4 disperses by 3.6 eV in the plane.

(b) Primitive lattice vectors and reciprocal lattice vectors

For `ibrav = 4` (hexagonal) the INPUT_PW documentation [1] gives the Cartesian vectors $\vec{a}_1 = a(1,0,0)$, $\vec{a}_2 = a(-\frac{1}{2},\frac{\sqrt{3}}{2},0)$ and $\vec{a}_3 = a(0,0,\frac{c}{a})$. `A` and `C` are lengths in Å [1], so with $a = 2.504$ Å and $c = 20.0$ Å

$$\vec{a}_1 = (2.504,\ 0,\ 0)\ \text{Å}, \qquad \vec{a}_2 = (-1.252,\ 2.169,\ 0)\ \text{Å}, \qquad \vec{a}_3 = (0,\ 0,\ 20.0)\ \text{Å}.$$

The reciprocal vectors follow from $\vec{a}_i \cdot \vec{b}_j = 2\pi\delta_{ij}$, i.e. $\vec{b}_1 = 2\pi\,\vec{a}_2 \times \vec{a}_3 / V$ and cyclic permutations, with $V = \vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)$ (Lecture 4):

$$\vec{a}_2 \times \vec{a}_3 = ac\,(\tfrac{\sqrt{3}}{2},\tfrac{1}{2},0), \quad \vec{a}_3 \times \vec{a}_1 = ac\,(0,1,0), \quad \vec{a}_1 \times \vec{a}_2 = a^2\,(0,0,\tfrac{\sqrt{3}}{2}), \quad V = \tfrac{\sqrt{3}}{2}a^2c = 108.60\,\text{Å}^3,$$

$$\vec{b}_1 = \frac{2\pi}{a}\Big(1,\tfrac{1}{\sqrt{3}},0\Big), \qquad \vec{b}_2 = \frac{2\pi}{a}\Big(0,\tfrac{2}{\sqrt{3}},0\Big), \qquad \vec{b}_3 = \frac{2\pi}{c}\,(0,0,1).$$

With $2\pi/a = 2.5093$/Å and $2\pi/c = 0.314\,16$/Å, the numerical values are in Table 1. $|\vec{b}_1| = |\vec{b}_2| = 4\pi/(\sqrt{3}a) = 2.8974$/Å, and the two vectors are at 60° to each other, so the reciprocal lattice is

[1] Inputs, outputs and analysis scripts (folder `runs/` on branch `claude/fervent-ptolemy-sry1eq`): `https://github.com/01nietzsch/dfp-ps2`

hexagonal. As a check, $\vec{a}_2 \cdot \vec{b}_1 = 2\pi(-\frac{1}{2} + \frac{\sqrt{3}}{2}\frac{1}{\sqrt{3}}) = 0$ and $\vec{a}_2 \cdot \vec{b}_2 = 2\pi$; numerically $\vec{a}_i \cdot \vec{b}_j = 2\pi\delta_{ij}$ to 10^{-15}. In units of $2\pi/a$ the vectors are $(1, 0.57735, 0)$, $(0, 1.15470, 0)$ and $(0, 0, 0.12520)$, identical to the reciprocal axes printed by `pw.x`.

Table 1: Primitive reciprocal lattice vectors (b) and high-symmetry points (c): crystal coordinates (coefficients of $\vec{b}_1, \vec{b}_2, \vec{b}_3$), Cartesian components and length.

	crystal coord.	k_x (1/Å)	k_y (1/Å)	k_z (1/Å)	length (1/Å)
$\vec{b}_1$	$(1, 0, 0)$	2.5093	1.4487	0	2.8974
$\vec{b}_2$	$(0, 1, 0)$	0	2.8974	0	2.8974
$\vec{b}_3$	$(0, 0, 1)$	0	0	0.3142	0.3142
Γ	$(0, 0, 0)$	0	0	0	0
K	$(\frac{1}{3}, \frac{1}{3}, 0)$	0.8364	1.4487	0	1.6728
M	$(\frac{1}{2}, 0, 0)$	1.2546	0.7244	0	1.4487

(c) K and M in terms of the primitive reciprocal lattice vectors

The first Brillouin zone is the region closer to $\vec{G} = 0$ than to any other $\vec{G}$ (Lecture 4). Its boundary is formed by the perpendicular bisectors of the six shortest $\vec{G}$, namely $\pm\vec{b}_1$, $\pm\vec{b}_2$ and $\pm(\vec{b}_2 - \vec{b}_1)$, which all have length $|\vec{b}_1|$.

M **(middle of a side).** The side belonging to $\vec{b}_1$ is its perpendicular bisector, and the midpoint of that side is where it crosses the line from Γ to $\vec{b}_1$. So $\vec{k}_M = \frac{1}{2}\vec{b}_1$, i.e. crystal coordinates $(\frac{1}{2}, 0, 0)$.

K **(corner).** The sides belonging to $\vec{b}_1$ and $\vec{b}_2$ meet at a corner, which is therefore equidistant from Γ, $\vec{b}_1$ and $\vec{b}_2$. Write $\vec{k}_K = x\vec{b}_1 + y\vec{b}_2$ and use $|\vec{b}_1|^2 = |\vec{b}_2|^2 \equiv B^2$ and $\vec{b}_1 \cdot \vec{b}_2 = B^2/2$:

$$|\vec{k}_K| = |\vec{k}_K - \vec{b}_1| \Rightarrow 2\vec{k}_K \cdot \vec{b}_1 = B^2 \Rightarrow 2x + y = 1, \qquad |\vec{k}_K| = |\vec{k}_K - \vec{b}_2| \Rightarrow x + 2y = 1,$$

so $x = y = \frac{1}{3}$. Hence $\vec{k}_K = \frac{1}{3}\vec{b}_1 + \frac{1}{3}\vec{b}_2$, crystal coordinates $(\frac{1}{3}, \frac{1}{3}, 0)$; Γ is $(0, 0, 0)$.

Check in Cartesian coordinates. $\vec{k}_K = \frac{2\pi}{a}(\frac{1}{3}, \frac{1}{\sqrt{3}}, 0)$ and $\vec{k}_M = \frac{2\pi}{a}(\frac{1}{2}, \frac{1}{2\sqrt{3}}, 0)$ (Table 1). These give $|\Gamma K| = 4\pi/(3a) = 1.6728/\text{Å}$, $|\Gamma M| = 2\pi/(\sqrt{3}a) = 1.4487/\text{Å}$ and $|KM| = 2\pi/(3a) = 0.8364/\text{Å}$. In a regular hexagon the distance to a corner equals the side length s, and the distance to the middle of a side is $\frac{\sqrt{3}}{2}s$. Here $|\Gamma M|/|\Gamma K| = \sqrt{3}/2$ and $|KM| = \frac{1}{2}|\Gamma K|$, so M is the midpoint of the side that starts at K and runs to $K' = (\frac{2}{3}, -\frac{1}{3}, 0)$, as in the sketch on the problem sheet. A numerical search over $\vec{G}$ also confirms that K is equidistant from its 3 nearest $\vec{G}$ (a corner) and M from its 2 nearest (a side midpoint).

(d) Band structure of monolayer hBN

The pseudopotentials have $Z_{\text{val}} = 3$ (B) and 5 (N), so there are 8 valence electrons per cell. In a spin-unpolarised calculation each band holds 2 electrons, so bands 1–4 are occupied and bands 5–8 are empty (Fig. 1). The valence-band maximum (top of band 4) and the conduction-band minimum (bottom of band 5) are both at K (Table 2). **hBN therefore has a direct band gap at K,** $E_g = \varepsilon_{5,K} - \varepsilon_{4,K} = \mathbf{0.207\,eV} - (\mathbf{-4.343\,eV}) = \mathbf{4.55\,eV}$**.** A path three times denser (271 k-points) and a fine path within 0.17/Å of K (201 k-points) give the same location and the same gap.

Table 2: Valence (band 4) and conduction (band 5) band edges at the high-symmetry points, relative to the VBM. The band extrema are in bold.

	band 4 (eV)	band 5 (eV)	direct gap (eV)
Γ	−1.398	4.793	6.191
K	**0.000**	**4.550**	**4.550**
M	−0.928	4.651	5.579

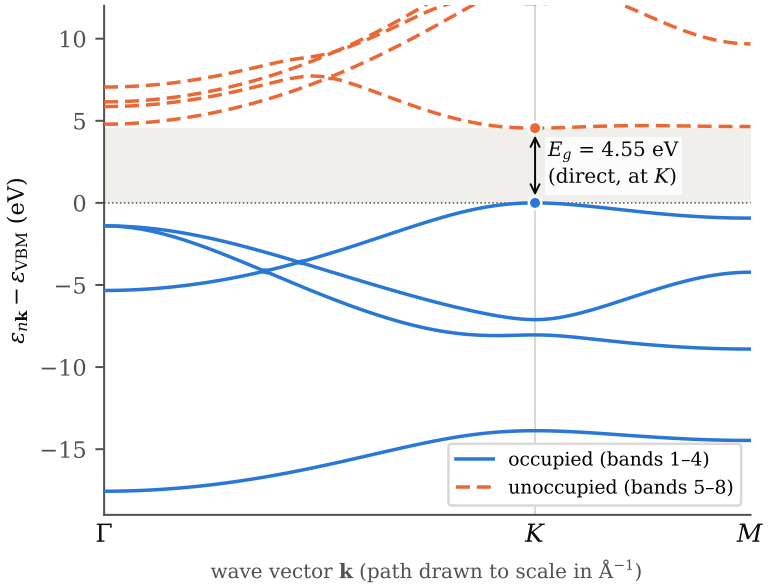


Figure 1: LDA Kohn–Sham band structure of monolayer hBN along Γ–K–M (path to scale: $|\Gamma K| = 1.673/\text{Å}$, $|KM| = 0.836/\text{Å}$). Energies are relative to the valence-band maximum. Solid blue: occupied bands 1–4. Dashed orange: unoccupied bands 5–8. Shading: band gap.

Table 3: Band gap of monolayer hBN compared with the experimental value 6.1 eV [2].

Method	E_g	error vs. 6.1 eV
LDA (`pz-vbc`, this work)	4.55 eV, direct at K	−1.55 eV (−25 %)
Experiment [2]	6.1 eV, direct	—

Comparison with experiment. The measured gap of monolayer hBN is 6.1 eV, and it is also direct [2]. LDA reproduces the direct character but underestimates the gap by 1.55 eV (about 25 %, Table 3). This is the known failure discussed in Lecture 3. LDA and GGAs work well for structural, elastic and vibrational properties but are inaccurate for band gaps of solids, and semi-local functionals systematically underestimate them. Two reasons apply here. First, the KS energies are eigenvalues of the fictitious non-interacting system, not true excitation energies (Lecture 2). Second, LDA takes ϵ_{xc} from the homogeneous electron gas, evaluated locally. Its approximate exchange does not cancel the spurious interaction of each electron with its own density, which is contained in the Hartree term; exact (Hartree–Fock) exchange does cancel it. Hybrid functionals such as PBE0 replace 25 % of the semi-local exchange by Hartree–Fock exchange, which opens the gap towards experiment, but they are roughly an order of magnitude more expensive.

References

[1] Quantum ESPRESSO, *Input File Description, Program pw.x* (INPUT_PW): entries `ibrav`, `A/C`, `K_POINTS`; `https://www.quantum-espresso.org/Doc/INPUT_PW.html` (accessed 10 Nov 2026).
[2] C. Elias, P. Valvin, T. Pelini, A. Summerfield, C. J. Mellor, T. S. Cheng, L. Eaves, C. T. Foxon, P. H. Beton, S. V. Novikov, B. Gil, G. Cassabois, "Direct band-gap crossover in epitaxial monolayer boron nitride", *Nat. Commun.* **10**, 2639 (2019), doi:10.1038/s41467-019-10610-5.
[3] P. Giannozzi *et al.*, *J. Phys.: Condens. Matter* **29**, 465901 (2017).